



DATA ANALYTICS TRACK • GRADUATION PROJECT

PHARMACEUTICAL WASTE & PROFIT LEAKAGE ANALYSIS

Where Data Meets the Pharmacy Supply Chain

GRADUATION PROJECT DOCUMENTATION

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Engineering

Technical Architecture

1 Technical Architecture

This chapter documents the tools, architectural pattern, and data model underlying the project. The pipeline follows the Medallion Architecture and outputs a star-schema dimensional model that feeds both the PharmAnalytics dashboard and the risk-prediction model.

1.1 Tools Used

Tool / Platform	Role in Project	Key Usage
Python (Pandas)	Data engineering & transformation	Ingesting raw transaction records, cleaning, feature engineering, and building the Bronze / Silver / Gold tables.
Scikit-learn	Machine learning	Training and evaluating the Random Forest risk-classification model; label encoding and train/test splitting.
Jupyter / Google Colab	Development environment	Interactive notebook environment for the end-to-end ETL and modeling workflow, including an ipywidgets-based prediction simulator.
Figma	UI/UX design & dashboard prototyping	Designing the PharmAnalytics dark-theme dashboard — layout, navigation, color system, and iconography — before implementation.
Power BI / PharmAnalytics BI Platform	Business intelligence & interactive dashboard	Consuming the Gold-layer star schema to power the seven-page executive dashboard: KPI cards, bar/line/donut charts, and Pareto analysis.

The pipeline begins with raw CSV ingestion into a Bronze layer that preserves the source data unmodified. A Python-based transformation stage (Silver layer) applies text standardization and derives the financial KPIs described in Chapter 2. The Gold layer restructures the enriched table into a star schema, which is consumed directly by the dashboard and by the Random Forest risk model.

1.2 Architecture (Medallion)

The Medallion Architecture organizes the pipeline into three logical layers, ensuring raw data is never overwritten, all cleaning logic is centralized, and the analytical layer is purpose-built for reporting performance.

Layer	Purpose	Key Operations
Bronze	Raw ingestion	CSV import, no transformation — permanent unmodified archive of the 8 raw transaction fields.
Silver	Cleaning & feature engineering	Drug-name standardization; derivation of Revenue, Total_Cost, Expired_Loss, Damaged_Loss, Net_Profit, Waste_Quantity, Waste_Rate, Profit_Margin; date decomposition into Year / Month / Month_Name.
Gold	Star schema	Split into 1 fact table (fact_sales) and 3 dimension tables (dim_branch, dim_date, dim_product), joined via surrogate keys.

1.3 Data Model (Star Schema)

The Gold layer implements a star schema consisting of one central fact table surrounded by three dimension tables, connected via surrogate keys (BranchKey, DateKey, ProductKey).

Table	Type	Key Columns	Description
fact_sales	Fact	DateKey, ProductKey, BranchKey	Central table: Sales_Quantity, Unit_Price, Unit_Cost, Revenue, Total_Cost, Net_Profit, Profit_Margin, Expired_Quantity, Damaged_Quantity, Waste_Quantity, Waste_Rate
dim_branch	Dimension	BranchKey	Branch identifier lookup (Branch_01 – Branch_15)
dim_date	Dimension	DateKey	Date, Year, Month, MonthName, Quarter
dim_product	Dimension	ProductKey	Drug_Name lookup for all 50 products

All dimension keys are generated as sequential surrogate integers and joined onto the enriched Silver table to populate fact_sales, following the same surrogate-key pattern used in classic star-schema design: each fact row references exactly one branch, one date, and one product.

Data Engineering

2 Data Engineering

This chapter documents the transformation logic applied between the Bronze and Silver layers, and the composite KPI columns engineered for analysis and modeling.

2.1 Cleaning Process

The Bronze layer ingests the raw CSV export exactly as received, with no transformation, to preserve a permanent unmodified archive. The Silver layer then applies a single standardization pass: drug names are normalized to consistent title case (for example, “Paracetamol 500mg” in the raw export becomes “Paracetamol 500Mg” in the Silver table), so that the same product is never split across multiple label variants in downstream aggregations and dimension lookups.

Field	Issue Addressed	Correction Applied
Drug_Name	Inconsistent capitalization across source rows	Standardized to title case before being used as the dim_product natural key
Date	Stored as text in the raw export	Parsed to a date type and decomposed into Year, Month, and Month_Name in dim_date
Branch	Free-text label	Mapped to a surrogate BranchKey via dim_branch

2.2 Feature Engineering

Eight columns are engineered in the Silver layer from the four raw numeric fields (Sales_Quantity, Unit_Price, Unit_Cost, Expired_Quantity, Damaged_Quantity). These derived fields carry all of the financial and waste signal used throughout the dashboard and the machine learning model.

Derived Column	Formula	Purpose
Revenue	$\text{Sales_Quantity} \times \text{Unit_Price}$	Gross sales value of the transaction
Total_Cost	$\text{Sales_Quantity} \times \text{Unit_Cost}$	Procurement cost of the units sold
Expired_Loss	$\text{Expired_Quantity} \times \text{Unit_Cost}$	Value written off due to expiry
Damaged_Loss	$\text{Damaged_Quantity} \times \text{Unit_Cost}$	Value written off due to physical damage
Net_Profit	$\text{Revenue} - \text{Total_Cost} - \text{Expired_Loss} - \text{Damaged_Loss}$	Profit after accounting for both waste channels
Waste_Quantity	$\text{Expired_Quantity} + \text{Damaged_Quantity}$	Total units lost to waste
Waste_Rate	$\text{Waste_Quantity} \div \text{Sales_Quantity}$	Waste as a share of units sold
Profit_Margin	$\text{Net_Profit} \div \text{Revenue}$	Profitability of the transaction, net of waste

Illustrative transformation logic (Silver layer):

```
df["Revenue"] = df["Sales_Quantity"] * df["Unit_Price"]
df["Total_Cost"] = df["Sales_Quantity"] * df["Unit_Cost"]
df["Expired_Loss"] = df["Expired_Quantity"] * df["Unit_Cost"]
df["Damaged_Loss"] = df["Damaged_Quantity"] * df["Unit_Cost"]
```

```
df["Net_Profit"]    = (df["Revenue"] - df["Total_Cost"]  
                      - df["Expired_Loss"] - df["Damaged_Loss"])  
df["Waste_Quantity"] = df["Expired_Quantity"] + df["Damaged_Quantity"]  
df["Waste_Rate"]    = df["Waste_Quantity"] / df["Sales_Quantity"]  
df["Profit_Margin"] = df["Net_Profit"] / df["Revenue"]  
df["Year"]          = df["Date"].dt.year  
df["Month"]         = df["Date"].dt.month  
df["Month_Name"]    = df["Date"].dt.month_name()
```

No rows are dropped during Silver-layer processing — every one of the 821,250 Bronze rows carries through to the Gold fact table, with the eight derived columns appended. The Gold layer then joins surrogate keys from dim_branch, dim_date, and dim_product onto this enriched table to produce fact_sales.

Machine Learning

3 Machine Learning

Beyond descriptive dashboards, the project incorporates a supervised classification model that flags each individual transaction with a Risk_Level — Low, Medium, or High — so that operations teams can identify waste-and-profit risk before it materializes, rather than reading it off a report after the fact.

3.1 ML Model Overview

The target variable, Risk_Level, is not present in the raw data — it is constructed from a transparent, rule-based scoring system built directly on the Silver-layer financial fields. Each transaction accumulates points based on four risk signals, and the accumulated score determines its risk tier:

Condition	Points Added
Waste_Rate $\geq$ 0.10 (10% of units sold lost to waste)	+2
Expired_Quantity $\geq$ 3 units	+2
Damaged_Quantity $\geq$ 2 units	+1
Profit_Margin < 0.20 (below 20% margin)	+1

Total Score	Assigned Risk_Level
4 or higher	High Risk
2 – 3	Medium Risk
0 – 1	Low Risk

Rule-based labeling logic:

```
def risk_level(row):
    score = 0
    if row["Waste_Rate"] >= 0.10:      score += 2
    if row["Expired_Quantity"] >= 3:   score += 2
    if row["Damaged_Quantity"] >= 2:   score += 1
    if row["Profit_Margin"] < 0.20:    score += 1

    if score >= 4: return "High Risk"
    elif score >= 2: return "Medium Risk"
    else:         return "Low Risk"

df["Risk_Level"] = df.apply(risk_level, axis=1)
```

Applying this rule across all 821,250 transactions produces a highly imbalanced but realistic class distribution — the overwhelming majority of pharmacy transactions carry no meaningful waste or margin risk:

Risk_Level	Transaction Count	Share of Dataset
Low Risk	755,445	91.97%
Medium Risk	33,663	4.10%

Risk_Level	Transaction Count	Share of Dataset
High Risk	32,142	3.91%

3.2 Model Training & Results

A Random Forest Classifier was trained to predict Risk_Level directly from operational fields available at the point of sale — without using the waste and margin fields that construct the label itself, so the model must learn the risk pattern from product, branch, and transaction-value signals alone.

Parameter	Value
Algorithm	Random Forest Classifier (scikit-learn)
Trees (n_estimators)	200
Random State	42
Features	ProductKey, BranchKey, Sales_Quantity, Revenue, Total_Cost
Target	Risk_Level (label-encoded: High / Medium / Low)
Train / Test Split	80% / 20% — 657,000 training rows / 164,250 test rows

Training and evaluation logic:

```

features = ["ProductKey", "BranchKey", "Sales_Quantity", "Revenue", "Total_Cost"]
X = df[features]
Y = LabelEncoder().fit_transform(df["Risk_Level"])

X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=42)

model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, Y_train)
predictions = model.predict(X_test)

acc = accuracy_score(Y_test, predictions)  # 0.9850

```

The model achieved an overall accuracy of 98.50% on the held-out test set. Per-class performance shows the expected pattern for an imbalanced target — near-perfect on the dominant Low Risk class, and strong, though slightly lower, performance on the two minority risk tiers:

Risk_Level	Precision	Recall	F1-Score	Support
High Risk	0.85	0.85	0.85	6,388
Low Risk	1.00	1.00	1.00	151,196
Medium Risk	0.84	0.84	0.84	6,666
Accuracy			0.9850	164,250
Macro Avg	0.90	0.90	0.90	164,250
Weighted Avg	0.99	0.98	0.98	164,250

3.3 Why Random Forest Was Selected

Random Forest was used as the production classifier for this task for several practical reasons:

- Non-linear, rule-derived target: because Risk_Level is built from threshold-based conditions (Waste_Rate $\geq$ 0.10, Expired_Quantity $\geq$ 3, etc.), the underlying decision boundaries are naturally step-like — exactly the pattern a tree-based ensemble captures well without manual feature transformation.
- Robustness to class imbalance: with Low Risk transactions outnumbering High Risk transactions more than 23 to 1, an ensemble of decision trees remains stable and interpretable without requiring aggressive rebalancing, though further gains are possible with class-weighting or oversampling (noted below as future work).
- No feature scaling required: ProductKey, BranchKey, Sales_Quantity, Revenue, and Total_Cost sit on very different numeric scales; Random Forest splits on raw feature values and needs no StandardScaler step, simplifying the production pipeline.
- Strong out-of-the-box performance: with default-adjacent hyperparameters (200 trees) and no extensive tuning, the model reached 98.5% accuracy and a 0.90 macro F1-score, indicating the five input features carry a genuinely predictive signal for waste-and-profit risk.

Future work: replacing the current majority-class-favoring split with a stratified or SMOTE-based resampling strategy, and adding Waste_Rate history or seasonality features, could further close the precision/recall gap on the Medium and High Risk tiers.

3.4 Confusion Matrix & Error Analysis

The confusion matrix below (rows = actual class, columns = predicted class, in the order High Risk, Low Risk, Medium Risk) shows where the model's errors concentrate:

Actual \ Predicted	High Risk	Low Risk	Medium Risk
High Risk	5,459	182	747
Low Risk	166	150,693	337
Medium Risk	767	269	5,630

Misclassifications concentrate almost entirely between the High Risk and Medium Risk tiers — both boundary conditions in the original scoring rule (a score of 3 vs. 4, for example) — while confusion with the dominant Low Risk class is minimal in both directions. This confirms the model is struggling with genuinely ambiguous, borderline-risk transactions rather than making gross errors.

3.5 Interactive Risk Prediction Tool

To make the model usable outside of a static report, an interactive prediction widget was built directly in the notebook using ipywidgets. A user enters a Product key, Branch key, Sales Quantity, Revenue, and Total Cost for a hypothetical or upcoming transaction, and the trained Random Forest model returns a predicted Risk_Level on demand — giving purchasing and operations staff a lightweight what-if tool for flagging risky stock decisions before they happen.

```
product = widgets.IntText(description='Product')
```

```
branch = widgets.IntText(description='Branch')
sales = widgets.FloatText(description='Sales')
revenue = widgets.FloatText(description='Revenue')
cost = widgets.FloatText(description='Cost')

def predict(b):
    user = pd.DataFrame({"ProductKey": [product.value],
                        "BranchKey": [branch.value],
                        "Sales_Quantity": [sales.value],
                        "Revenue": [revenue.value],
                        "Total_Cost": [cost.value]})

    pred = model.predict(user)
    risk = encoder.inverse_transform(pred)
    print("Predicted Risk Level:", risk[0])

button.on_click(predict)
```

This closes the loop between descriptive dashboard insight and predictive analytics: the PharmAnalytics dashboard shows where waste and profit leakage happened, while the Random Forest model — through this simulator — estimates where it is likely to happen next.

Analysis

CHAPTER 4

Project Overview

4 Project Overview

Pharmaceutical retail sits at the intersection of tight margins and perishable inventory: every expired vial of insulin or damaged box of antibiotics is a direct hit to profitability, not just a shrinkage statistic. As multi-branch pharmacy networks scale, the ability to see — branch by branch, drug by drug, month by month — exactly where revenue is being generated and where it is leaking away becomes a core business intelligence requirement.

4.1 Project Concept

Pharmaceutical Waste & Profit Leakage Analysis is a graduation project developed for the Data Analytics Track. The project analyzes daily, branch-level pharmaceutical sales transactions — spanning 13 active branches and a 50-drug portfolio — to quantify how much revenue is generated, how much profit is retained, and how much value is lost to expired and damaged stock across the network.

The data pipeline was built following the Medallion Architecture (Bronze – Silver – Gold), transforming raw transactional records into a star-schema analytical model. The project culminates in “PharmAnalytics” — a seven-page executive Business Intelligence dashboard, prototyped first in Figma and then implemented as an interactive reporting platform — alongside a supervised machine learning model that flags individual transactions as Low, Medium, or High waste-and-profit risk.

The motivation behind this project is twofold:

- Analytical: apply the complete data analysis pipeline — from raw ingestion and cleaning through dimensional modeling, predictive modeling, and interactive visualization — to a real, financially meaningful operations domain.
- Operational: surface which drugs, branches, and months carry the highest waste and profit-leakage risk, giving pharmacy operations teams a concrete basis for stock rotation, purchasing, and intervention decisions.

Project objectives span five domains:

- Data Engineering: ingest and clean 821,250 branch-level transaction records across the Medallion Architecture layers.
- Feature Engineering: derive financial KPIs — Revenue, Net Profit, Waste Quantity, Waste Rate, and Profit Margin — from raw pricing and quantity fields.
- Dimensional Modeling: design a star-schema Gold layer with a central fact table and three supporting dimensions (Branch, Date, Product), optimized for BI consumption.
- Predictive Modeling: train a classification model that predicts a per-transaction Risk_Level, enabling proactive rather than reactive waste management.
- Dashboard Development: build a seven-page interactive dashboard covering profitability, waste & loss, branch performance, inventory, and executive summary views.

4.2 Data Source

The dataset is a synthetic, transaction-level pharmaceutical retail dataset capturing daily sales, pricing, and stock-loss records for every drug at every branch.

Attribute	Detail
Dataset Type	Branch-level daily pharmaceutical sales transactions (synthetic)

Attribute	Detail
Granularity	One row per Branch × Drug × Day
Total Branches	13 active branches (15 branch keys defined in the dimension table)
Total Products	50 distinct drugs / medical products
Total Rows (Fact Table)	821,250 transaction records
Reporting Period	January 2021 onward — dashboard default view: Jan – Dec 2024
File Format	CSV, transformed to structured Bronze / Silver / Gold tables
Currency	Egyptian Pound (EGP)

The raw (Bronze) dataset contains 8 columns spanning transaction identifiers, pricing, and stock-loss counts:

Column Name	Type	Description
Date	Date	Calendar date of the transaction
Branch	String	Branch identifier (Branch_01 – Branch_15)
Drug_Name	String	Name and dosage of the drug or medical product
Sales_Quantity	Integer	Units sold on that day at that branch
Unit_Price	Float	Retail selling price per unit (EGP)
Unit_Cost	Float	Procurement cost per unit (EGP)
Expired_Quantity	Integer	Units removed from stock due to expiry
Damaged_Quantity	Integer	Units removed from stock due to physical damage

4.3 Project Impact (KPIs)

A set of engineered financial and operational KPIs is derived in the Silver layer and consumed throughout the PharmAnalytics dashboard.

Profitability KPIs

KPI	Definition	Value
Total Revenue	Sum of Sales_Quantity × Unit_Price across all transactions	2.69bn EGP
Total Net Profit	Revenue minus cost, expired loss, and damaged loss	982.62M EGP
Average Profit Margin	Net Profit ÷ Revenue, averaged across the portfolio	36.49%
Highest Revenue Product	Drug with the largest total revenue contribution	Cough Syrup
Highest Margin Product	Drug with the largest average profit margin	Vitamin D3
Loss-Making Drugs	Drugs with negative aggregate Net Profit	3 drugs

Waste & Loss KPIs

KPI	Definition	Value
Total Waste (Cost)	Sum of Expired_Loss + Damaged_Loss across all transactions	26.96M EGP

KPI	Definition	Value
Expired Loss	Expired_Quantity × Unit_Cost, summed	16.78M EGP
Damaged Loss	Damaged_Quantity × Unit_Cost, summed	10.18M EGP
Waste % of Revenue	Total Waste ÷ Total Revenue	1.00%
Waste % of Cost	Total Waste ÷ Total Cost of Goods	1.58%
Profit Lost to Waste	Waste value as a share of potential (waste-free) profit	2.74%
Highest Waste Month	Calendar month with the largest waste value	January

Branch Performance KPIs

KPI	Definition	Value
Total Branches	Count of branches with reported activity	13
Top Branch	Branch with the highest total Net Profit	Branch_03 (91M EGP)
Average Net Profit / Branch	Total Net Profit ÷ number of branches	69.5M EGP
Network Total Net Profit	Sum of Net Profit across all branches	903.5M EGP
Lowest Branch	Branch with the smallest total Net Profit	Branch_09 (55M EGP)

Inventory KPIs

KPI	Definition	Value
Total Inventory Cost	Cumulative procurement value of goods sold	1.68bn EGP
Dead Stock Rate	Share of products with negligible sales movement	0.81%
A-Class Products	High-value products (top tier of ABC / Pareto analysis)	8 products
B-Class Products	Medium-value products	24 products
C-Class Products	Low-value, slow-moving products	18 products

4.4 Main Charts Overview

The seven-page PharmAnalytics dashboard follows a data-storytelling structure — opening with project context, moving through profitability, waste, branch, and inventory deep-dives, and closing with an executive roll-up and the underlying design system.

Page	Title	Focus Area	Key KPIs Shown
1	Landing	Project identity, team, and headline scale metrics	Total Branches, Total Products, Total Revenue, Profit Margin
2	Profitability	Revenue, net profit, and margin by drug	Highest Revenue Product, Total Revenue, Total Net Profit, Avg. Profit Margin
3	Waste & Loss	Expired and damaged stock cost	Total Waste, Expired Loss, Damaged Loss, Waste % of Revenue
4	Branch Performance	Branch-level profit and waste comparison	Total Branches, Top Branch, Avg. Net Profit, Network Net Profit

Page	Title	Focus Area	Key KPIs Shown
5	Inventory	Dead stock and ABC / Pareto classification	Total Inventory Cost, Dead Stock Rate, A/B/C-Class Products
6	Executive Summary	CEO-level roll-up across all dimensions	Total Revenue, Total Net Profit, Total Waste, Revenue Growth
7	Style Guide	Design system reference (colors and icons)	Color Palette, Icon Library

4.5 SMART Questions

The analysis is guided by five SMART (Specific, Measurable, Achievable, Relevant, Time-bound) research questions:

#	SMART Question	KPI / Measure
1	Does total pharmaceutical waste exceed 1% of total revenue across the network, and which loss type — expired or damaged — dominates?	Total Waste, Waste % of Revenue, Expired vs Damaged Loss
2	How much does Net Profit vary between the top-performing and lowest-performing branches, and by what margin?	Net Profit by Branch, Top Branch, Lowest Branch
3	Do insulin products (Glargine, Aspart, Regular) show waste rates above 15%, and what share of total waste cost do they represent?	Waste Rate by Drug, Top Contributors to Waste Cost
4	What proportion of the 50-drug portfolio qualifies as Class A inventory, and does dead stock concentrate among low-turnover drugs?	ABC Distribution, Dead Stock Rate
5	How closely does actual monthly profit track potential (waste-free) profit, and how much profit is recovered vs. permanently lost each month?	Potential Profit vs Actual Profit, Profit Lost to Waste

CHAPTER 5

Dashboard & Visualization

5 Dashboard & Visualization

This chapter presents the final end-user deliverable of the project — “PharmAnalytics,” a seven-page interactive Business Intelligence dashboard that turns 821,250 transaction rows into a navigable, executive-ready data story. The chapter covers the page inventory, headline KPI reference, the underlying design system, and a page-by-page walkthrough.

5.1 PharmAnalytics Dashboard Pages

The dashboard connects to the Gold-layer star schema and is organized into seven pages, accessible from a persistent left-hand navigation rail. It follows a deliberate data-storytelling structure — opening with project identity, then progressing through profitability, waste, branch, and inventory dimensions, before closing with an executive roll-up and the design system reference.

#	Page Title	Theme / Tagline
01	Landing	Pharmaceutical Waste and Profit Leakage Analysis System
02	Profitability	Revenue, Net Profit, Margin & Cost Structure
03	Waste & Loss	Expired Loss, Damaged Inventory & Waste Rates
04	Branch Performance	Branch-level Revenue, Profit & Operational Metrics
05	Inventory	Dead Stock, ABC Analysis & Sales Movement
06	Executive	Full Business Overview · CEO Decision Support
07	Style Guide	Color Palette & Icon Library

5.2 KPI Highlights

Across the six analytical pages, the following headline KPIs are surfaced as card visuals — each representing a critical dimension of pharmaceutical waste and profitability:

KPI	Value	Dashboard Page
Total Revenue	2.69bn EGP	Profitability / Executive
Total Net Profit	982.62M EGP	Profitability / Executive
Average Profit Margin	36.49%	Profitability / Executive
Highest Revenue Product	Cough Syrup	Profitability
Highest Margin Product	Vitamin D3	Profitability
Loss-Making Drugs	3	Profitability
Total Waste	26.96M EGP	Waste & Loss / Executive
Expired Loss	16.78M EGP	Waste & Loss
Damaged Loss	10.18M EGP	Waste & Loss
Waste % of Revenue	1.00%	Waste & Loss
Highest Waste Month	January	Waste & Loss
Total Branches	13	Branch Performance

KPI	Value	Dashboard Page
Top Branch	Branch_03 (91M EGP)	Branch Performance
Avg. Net Profit / Branch	69.5M EGP	Branch Performance
Network Net Profit	903.5M EGP	Branch Performance
Lowest Branch	Branch_09 (55M EGP)	Branch Performance
Total Inventory Cost	1.68bn EGP	Inventory
Dead Stock Rate	0.81%	Inventory
A / B / C-Class Products	8 / 24 / 18	Inventory
Revenue Growth	+8.40%	Executive
Waste % of Cost	1.58%	Executive
Profit Lost to Waste	2.74%	Executive

5.3 Design System & Figma

Before a single visual was placed in the reporting layer, the complete dashboard was designed and prototyped in Figma. This design-first approach let the team finalize the storytelling flow, dark-theme color palette, page layout, and iconography for every page — significantly reducing iteration time during implementation. The final Style Guide page (page 7) is embedded directly in the product as a living design-system reference.

Visual Identity

PharmAnalytics uses a deep-navy, indigo-accented dark theme — chosen to read as a modern, data-dense enterprise analytics product rather than a conventional light-mode report. Indigo and lavender accents distinguish primary metrics and active navigation states, cyan and sky blue highlight secondary chart series, and a restrained green / orange / red semantic set flags positive, cautionary, and negative KPI movement respectively.

Color Palette (as documented on the Style Guide page):

Color Name	Hex	Usage
Deep Navy	#0F172A	Page background
Navy Blue	#132347	Panel / card background
Midnight Blue	#182956	Card hover / secondary panel
Dark Indigo	#312E81	Chart secondary series
Indigo	#4F46E5	Primary accent — active nav, key bars
Lavender	#885CF6	Secondary accent
Soft Lavender	#C485FD	Tertiary accent / highlights
Light Lilac	#E9D5FF	Light accent text
Cyan Accent	#67E8F9	Line charts / trend series
Sky Blue	#38BDF8	Secondary line series
White	FFFFFF	Primary text on dark surfaces
Gray 200 / 400 / 500	#E5E7EB / #9CA3AF / #687280	Secondary text, borders, muted labels

Color Name	Hex	Usage
Success Green	#22C55E	Positive KPI movement
Warning Orange	#F59E0B	Caution / waste indicators
Danger Red	#EF4444	Negative KPI movement, loss-making items

An accompanying icon library (dashboard, revenue, profit, analytics, inventory, warehouse, medicine, flask, syringe, waste bin, warning, percent, branch, trophy, and more) gives every KPI card and navigation item a consistent, at-a-glance visual identity across all seven pages.

5.4 Dashboard Walkthrough

The following pages present each dashboard screen with its full screenshot and a description of the visuals and insights it delivers.

5.4.1 Page 1 — Landing

The landing page establishes the project's identity and scope. It displays the project title, “Executive Power BI Dashboard · Healthcare Business Intelligence Platform,” the Data Analytics Track badge, the project supervisor, and full project team — each member represented with a colored initial avatar and LinkedIn link. A row of headline scale metrics (branches, products, revenue, and margin) anchors the bottom of the page as the entry point into the rest of the dashboard.

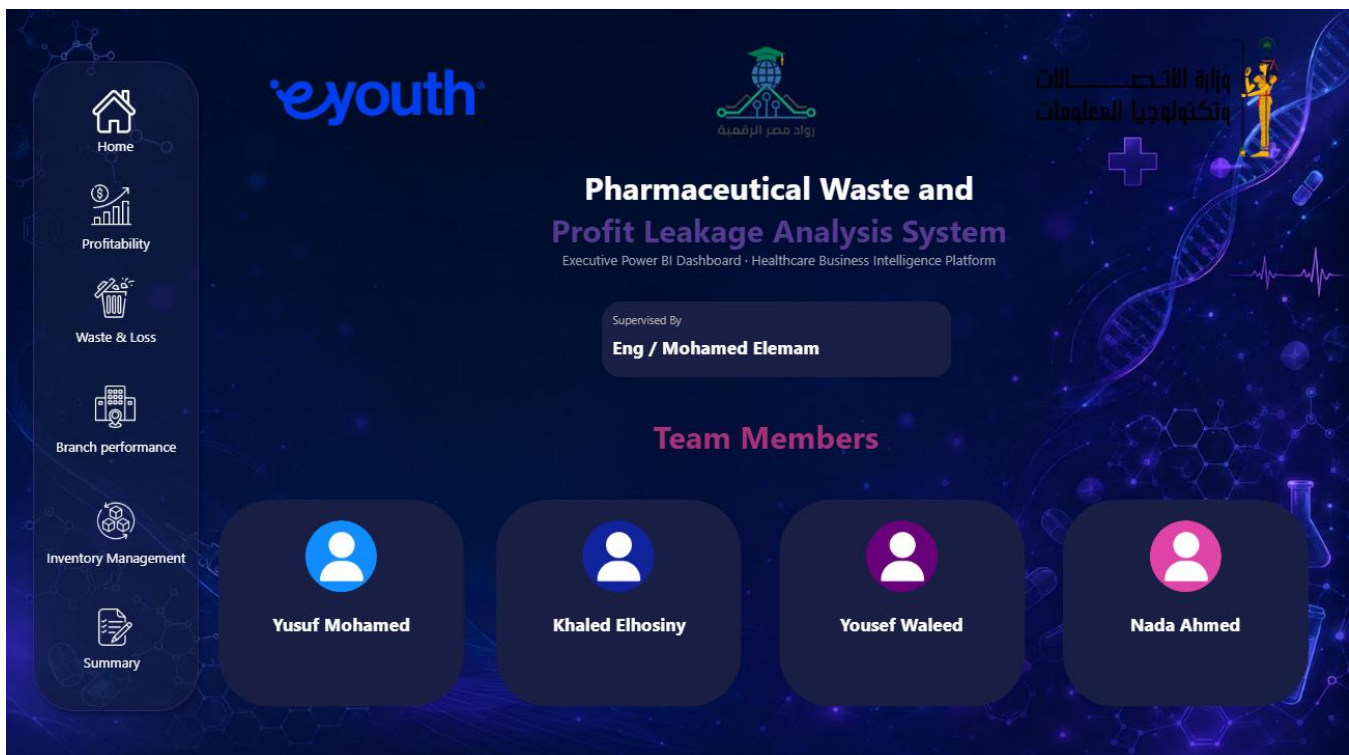


Figure 5.1 — Landing Page: project identity, team members, and headline scale metrics

5.4.2 Page 2 — Profitability & Financial Performance

The Profitability page breaks down revenue, net profit, and margin at the drug level. Six KPI cards headline the page, followed by a Revenue vs. Net Profit comparison, a donut chart of the top five profit contributors, a highlight of the

lowest-profitable (loss-making) drugs, a Top 10 Drugs by Net Profit ranking, and a monthly Cost Structure Analysis breaking spend into COGS, operating expenses, waste, and net profit.

Element	Type	Key Insight
Highest Revenue Product	KPI Card	Cough Syrup (+8.4%)
Total Revenue	KPI Card	2.69bn EGP (+5.2%)
Total Net Profit	KPI Card	982.62M EGP
Average Profit Margin	KPI Card	36.49%
Loss-Making Drugs	KPI Card	3 drugs with negative net profit
Revenue vs Net Profit by Drug	Bar Chart	Cough Syrup (465M) and ORS Sachets (425M) lead revenue
Drug Contributors to Total Profit	Donut Chart	Top 5 drugs contribute the large majority of total profit
Lowest Profitable Drugs	Bar Chart	Insulin Regular, Baby Formula and IV Cannula post negative net profit
Top 10 Drugs by Net Profit	Bar Chart	Cough Syrup (206M) and ORS Sachets (188M) are the top two profit generators
Cost Structure Analysis	Stacked Bar (Monthly)	COGS dominates the cost stack; waste remains a consistent, visible slice year-round



Figure 5.2 — Profitability & Financial Performance: KPI cards, Revenue vs Net Profit bar chart, Drug Contributors donut, and Cost Structure Analysis

5.4.3 Page 3 — Waste & Loss Analysis

The Waste & Loss page isolates the cost of expired and damaged stock. Five KPI cards summarize total waste, its two components, and its severity relative to revenue, followed by a ranked view of the drugs with the highest waste rate, a donut of top waste-cost contributors, and a comparison of expired versus damaged loss by drug.

Element	Type	Key Insight
Total Waste	KPI Card	26.96M EGP (-3.2% vs. prior period)
Expired Loss	KPI Card	16.78M EGP
Damaged Loss	KPI Card	10.18M EGP
Waste % of Revenue	KPI Card	1.00%
Highest Waste Month	KPI Card	January
Top Drugs by Waste Rate	Bar Chart (%)	Ceftriaxone 1G (18.51%), Ins. Glargine (18.19%), and Ins. Aspart (18.18%) lead by a wide margin
Top Contributors to Waste Cost	Donut Chart	Four cold-chain / injectable products (Ceftriaxone, Ins. Glargine, Ins. Aspart, Ins. Regular) drive ~96% of waste cost
Expired Loss by Drug	Bar Chart	Concentrated in the same high-waste-rate products
Expired vs Damaged Loss Distribution	Bar Chart	Insulin Regular and Ceftriaxone 1G show the largest combined loss bars



Figure 5.3 — Waste & Loss Analysis: KPI cards, Top Drugs by Waste Rate, Top Contributors to Waste Cost donut, and Expired vs Damaged Loss Distribution

5.4.4 Page 4 — Branch Performance Analysis

The Branch Performance page ranks all 13 active branches by profitability and waste. Five KPI cards headline network scale and the top/bottom performing branches, followed by a full branch ranking, a revenue-vs-profit comparison for the top six branches, a monthly trend line for the top three, and a waste comparison across branches.

Element	Type	Key Insight
Total Branches	KPI Card	13
Top Branch	KPI Card	Branch_03 (+12.4%, 91M EGP net profit)

Element	Type	Key Insight
Avg. Net Profit / Branch	KPI Card	69.5M EGP
Network Net Profit	KPI Card	903.5M EGP (+6.8%)
Lowest Branch	KPI Card	Branch_09 (55M EGP)
Net Profit by Branch (All 13)	Bar Chart	Branch_03 (91M), Branch_04 (85M), and Branch_01 (84M) form a clear top tier
Revenue vs Net Profit by Branch (Top 6)	Bar Chart	Branch_03 leads on both revenue and profit
Monthly Net Profit Trend — Top 3 Branches	Line Chart	Profit peaks mid-year and softens toward Q4 across the leading branches
Branch Waste Comparison	Bar Chart	Branch_01 (2.8M) shows the highest waste among the top branches despite strong profit



Figure 5.4 — Branch Performance Analysis: KPI cards, Net Profit by Branch ranking, Revenue vs Net Profit (Top 6), and Branch Waste Comparison

5.4.5 Page 5 — Inventory & Operational Insights

The Inventory page applies dead-stock identification and ABC (Pareto) classification across the 50-product portfolio. Five KPI cards summarize inventory value, dead stock, and the A/B/C tier split, followed by a dead-stock ranking, a monthly sales-quantity trend, a best-selling-drug-by-month table, and the full ABC Pareto chart.

Element	Type	Key Insight
Total Inventory Cost	KPI Card	1.68bn EGP (+4.1%)
Dead Stock Rate	KPI Card	0.81%
A-Class Products	KPI Card	8 high-value products

Element	Type	Key Insight
B-Class Products	KPI Card	24 medium-value products
C-Class Products	KPI Card	18 low-value products
Dead Stock Identification	Bar Chart (%)	Ceftriaxone 1G (19%) and the insulin family lead dead-stock exposure
Monthly Sales Quantity Trend	Area Chart	Sales dip mid-year and rebound sharply toward December
Monthly Best Selling Drug	Table	Cough Syrup leads Jan–Feb, Vitamin D3 1000 IU leads Mar–May, ORS Sachets leads Jun–Aug
ABC Distribution — Pareto Analysis	Bar + Line	The top ~8 products cross the 80% A/B boundary, confirming a classic Pareto concentration



Figure 5.5 — Inventory & Operational Insights: KPI cards, Dead Stock Identification, Monthly Sales Quantity Trend, and ABC Pareto Analysis

5.4.6 Page 6 — Executive Summary

The Executive Summary page is the dashboard's CEO-facing roll-up, consolidating profitability, waste, branch, and trend signals into a single decision-support view. Seven KPI cards headline the page, backed by an Expired-vs-Damaged donut, a Potential-vs-Actual monthly profit comparison, top net-profit and waste breakdowns, and a full-year monthly sales trend.

Element	Type	Key Insight
Total Revenue	KPI Card	2.69bn EGP (+8.4%)
Total Net Profit	KPI Card	982.62M EGP (+5.2%)
Total Waste	KPI Card	26.96M EGP (-3.2%)
Profit Lost to Waste	KPI Card	2.74%

Element	Type	Key Insight
Revenue Growth	KPI Card	+8.40%
Waste % of Cost	KPI Card	1.58%
Profit Margin	KPI Card	36.49%
Expired vs Damaged Loss	Donut Chart	Expired loss accounts for the larger share of total waste value
Potential Profit vs Actual Profit	Bar Chart (Monthly)	Actual profit consistently trails potential (waste-free) profit by a small, stable margin each month
Top Net Profit by Drug	Bar Chart	Cough Syrup and ORS Sachets remain the two largest profit contributors network-wide
Total Waste by Drug	Bar Chart	Insulin Glargine and Insulin Regular dominate network waste cost
Monthly Sales Trend	Line Chart	Mirrors the mid-year dip and December rebound seen on the Inventory page



Figure 5.6 — Executive Summary: KPI cards, Expired vs Damaged Loss donut, Potential vs Actual Profit, and Total Waste by Drug

5.4.7 Page 7 — Style Guide

Unlike the six analytical pages, the Style Guide page carries no business KPIs. It documents the dashboard's own design system — the full color palette and icon library — directly inside the product, so the visual language stays consistent as new pages or visuals are added in future iterations.

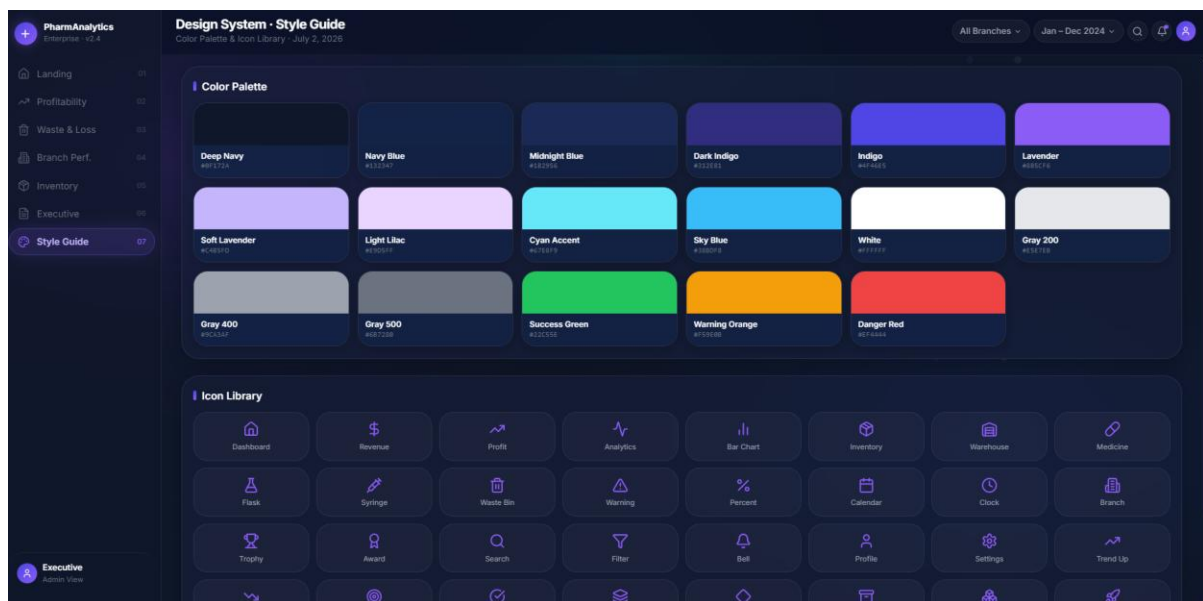


Figure 5.7 — Style Guide: Color Palette and Icon Library reference

5.5 Design Summary

The PharmAnalytics dashboard represents the convergence of a clean star-schema data model, a rule-based predictive layer, and intentional dark-theme design. Every element — from the indigo accent system to the KPI cards and Pareto analysis — was chosen to make 821,250 transaction rows accessible, scannable, and actionable for pharmacy operations decision-makers.

Design Dimension	Choice Made	Rationale
Background Theme	Deep navy / indigo dark theme across all seven pages	Reads as a modern, data-dense enterprise analytics product; reduces eye strain for extended executive review sessions
Navigation	Persistent left rail with numbered pages (01–07)	Keeps orientation clear across a seven-page, multi-domain dashboard
Color Palette	Indigo / lavender primary accents, cyan / sky-blue for trend lines, green / orange / red semantic set	Separates primary metrics from secondary series while keeping positive / negative movement instantly legible
KPI Cards	Four to seven headline KPIs per page with directional trend arrows	Immediate executive-level insight; scannable at a glance before drilling into charts
Design-First Workflow	Full Figma prototype preceding implementation	Finalized storytelling flow, layout, and color system before build, reducing iteration time
Interactivity	All-Branches and date-range filters on every page	Enables both network-wide and branch-specific exploration from a single report